DecloudNet: Cross-Patch Consistency is a Nontrivial Problem for Thin Cloud Removal From Wide-Swath Multispectral Images

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***Abstract*— Cloud cover leads to great loss of spatial details in wide-swath multispectral images, and thus significantly affects their application value. Wide-swath images are huge in size and are usually cropped into patches before thin cloud removal. In addition, wide-swath images contain a rich variety of types and shapes of thin clouds, with each patch containing different clouds. However, most of the existing methods and datasets were primarily designed for natural image dehazing. These datasets had limited types and shapes of clouds. When applied to remote sensing image thin cloud removal task, these methods are unable to remove various types of clouds and lead to severe cross-patch color difference. To address this problem, a DecloudNet based on cross-patch consistency supervision was proposed. First, a multiscale cloud perception block (MCPB) with multisize convolutional kernels was proposed to enhance the network's capability to extract clouds feature of different sizes. Second, a cross-patch consistency supervision was designed to reduce the network's inconsistent cloud removal strength in different patches and remove cross-patch color difference when processing wide-swath images. Finally, a thin cloud simulation method based on Perlin noise, domain warping, and atmospheric scattering model was proposed to construct a high-quality declouding dataset containing clouds of multiple sizes and shapes, which can improve the performance of DecloudNet on different kinds of thin clouds. The DecloudNet and compared methods were tested for simulated thin cloud removal performance on images from QuickBird (QB), GaoFen-2 (GF2), and WorldView-2 (WV2) satellites, and for real thin cloud removal performance on wide-swath images from GaoFen-1 Wide Field of View (GF-1 WFV), GaoFen-1 (GF-1), and Earth-Observing-1 (EO-1) satellites. The experimental results demonstrated that DecloudNet outperformed the existing state-of-the-art (SOTA) methods. DecloudNet and cross-patch consistency supervision made it possible to perform thin cloud removal on wide-swath images of large size on most GPUs without worrying about graphics memory limitation. The source code and dataset are available at https://github.com/N1rv4n4/DecloudNetthe link.**

***Index Terms*— Cross-patch consistency, multispectral image, thin cloud removal, wide-swath.**

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I. Introduction

REMOTE sensing has enabled humans to observe the Earth's surface on a large scale, allowing us to plan land use [1], [2], [3], protect the ecological environment [4], [5], [6], predict natural disasters [7], [8], [9], and so on. Clear remote sensing images without cloud cover can significantly improve the accuracy of these tasks [10], [11], [12]. In recent years, an increasing number of remote sensing satellites have been launched into orbit [13], [14], [15]. This provides a wealth of data of remote sensing images. Nevertheless, a significant portion of remote sensing images are affected by cloud cover, making the details of ground objects difficult to recognize. Therefore, it is crucial to apply thin cloud removal, or declouding processing, to remote sensing images.

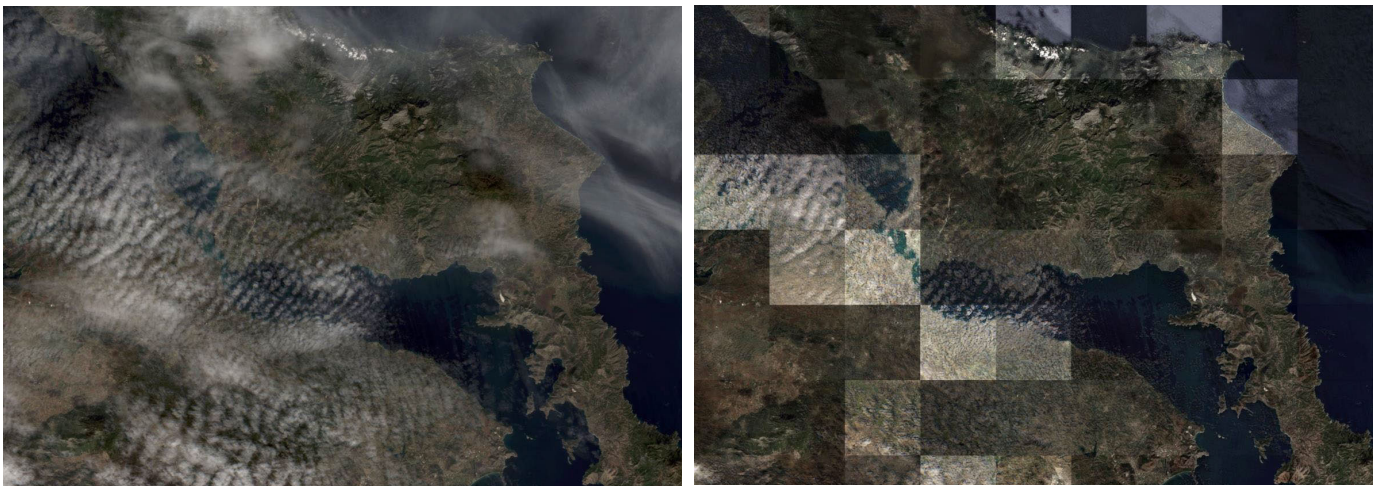
The existing dehazing and cloud removal tasks are similar, but not identical, to declouding. Dehazing was originally used to process fog in natural images [16], [17]. Natural images are taken at a different angle than remote sensing images. Moreover, the difference between fog in natural images and various clouds in remote sensing images is obvious. Therefore, the dehazing method is not entirely suitable for the task of thin cloud removal from remote sensing images. In contrast, most cloud removal methods are designed for remote sensing images [18], [19], [20]. However, cloud removal targets all kinds of clouds in remote sensing images, especially thick clouds that completely block ground objects. Therefore, it requires multitemporal remote sensing images to paste the ground objects in the cloudless image into the cloudy image. In addition, there are some cloud removal methods based on image fusion [21], [22], [23]. These methods use synthetic aperture radar (SAR) ability to see through clouds to recover cloud-obscured ground objects in multispectral images. In summary, these cloud removal methods require multitemporal or multisensor remote sensing images as input. They are not applicable to the case where a single multispectral image is used as input.

A. Dehazing Methods

Most of the existing dehazing methods are designed for fog in natural images. In feature fusion attention network (FFANet) [24], a novel feature attention module combined channel attention and pixel attention, considering that different channel features contained completely different weighting

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(a) (b)

Fig. 1. Thin cloud removal result of the existing methods for a cloud-covered wide-swath image of size 9000 × 6700. Due to the huge size of wide-swath images, the images need to be cropped and then fed into the deep network for processing. Existing methods perform well within one patch but suffer from color difference between patches. This demonstrates that the existing methods lack the ability to handle a variety of clouds and incorrectly change the pixel values of each patch. (a) Cloud covered input image. (b) Problem of existing methods.

information and haze was not uniformly distributed over different pixels. Pixel attention was an attention mechanism used to enhance the interaction between pixels in a deep network. It helped the network to focus more accurately on the important pixels in an image and adaptively adjust the weights of the pixels. Channel attention mechanism was an attention mechanism used to enhance the interaction between channels in a deep network. It helped the network to better learn the importance of features between channels and adaptively adjust the weights of the channels.

Three principles about building convolutional neural networks were proposed in universal perception large-kernel convNet (UniRepLKNet) [25]: use a small number of large convolutional kernels to ensure a large receptive field, use small convolutions such as depthwise to improve the feature abstraction level, and use some efficient structures to improve the depth of the model and thus enhance its general representation. These principles were also suitable for remote sensing image declouding tasks. The size and shape of clouds in remote sensing images were complex and variable. A combination of convolutional kernels of different sizes is required to enhance the network's ability to extract cloud features. Mix structure image dehazing network (MixDehazeNet) [26] used an encoder containing five different sized convolutional kernels to extract cloud features and proposed a simple pixel attention mechanism as a parallel branch to pixel attention and channel attention.

As for dehazing datasets [27], [28], [29], there were sufficient natural image dehazing datasets. However, the input images were made by adding simulated thin and uniformly distributed haze to the haze-free images. These simulated haze were quite different from the clouds in remote sensing images in terms of shape, thickness, and other features, which made it difficult to be used for training and testing of cloud removal methods for remote sensing images.

Most of the existing remote sensing image dehazing methods were modified from natural image dehazing methods. Modified methods based on dehazing have limited effectiveness in thin cloud removal when processing remote sensing wide-swath images. Meanwhile, the input image of the natural image dehazing network was much smaller than that of remote sensing image dehazing network. These methods could process a whole natural image at once but had to process remote sensing wide-swath images in patches. This results in severe cross-patch color difference in the results of thin cloud removal from remote sensing wide-swath images. The cross-patch color difference is shown in Fig. 1.

B. Cloud Removal Methods

Most of the existing cloud removal methods were designed for remote sensing images, especially those covered by thick clouds. To compensate for ground object information lost due to thick cloud cover, these methods use multisensor [30], [31] or multitemporal [32], [33], [34] image fusion. Some methods fused SAR images with multispectral images, and others fused near infrared (NIR) images with visible images. A deep residual network was proposed to remove clouds from multispectral Sentinel-2 images [30]. The recovery of the lost image details was accomplished by SAR and optical image fusion and evaluated by a new cloud-adaptive loss function. In cloud removal method for multispectral scanner image (CR-MSS) [31], a novel deep network that took visible/NIR bands, short-wave infrared bands, and vegetation red edge bands as inputs was proposed to remove clouds in Sentinel-2A images. It fully took advantage of the spectral information available in short-wave infrared bands and vegetation red edge bands. In [32], a novel spatiotemporal patch group deep learning network for gap-filling through multiple temporal cloudy images was proposed. It removed both thick clouds and cloud shadows, which significantly improved image quality.

These methods provided good cloud removal results, but the problem is that not all the multisensor remote sensing images are available for any region at any time. These methods are difficult to work when only multispectral images are

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available. As for the methods that fused multitemporal images to achieve cloud removal, they did not actually recover the ground information on the day of the cloudy weather, but used a replacement for when it was clear afterward. As a result, the timeliness of these methods was poor, making it difficult to provide a database for subsequent time-sensitive tasks such as target detection.

The existing remote sensing image cloud removal datasets [18], [35], [36], [37] included both the simulated and real datasets. These simulation datasets had the same problem as the natural image dehazing datasets, in which only thin and uniformly distributed fog was simulated. These real datasets mainly contained images with fog and thick clouds, but not enough variety in thin clouds.

C. Contributions

To address the problems of the existing dehazing or cloud removal methods and datasets, this article proposed the DecloudNet based on cross-patch consistency supervision and a simulation method that can simulate four common types of thin clouds. The contributions of this article are as follows.

1) A multiscale cloud perception block (MCPB) with multiscale convolutional kernels was proposed to enhance the network's capability to extract clouds feature of different sizes and shapes.
2) A cross-patch consistency supervision was designed to reduce the network's inconsistent declouding strength on different patches and remove cross-patch color difference when processing wide-swath images.
3) A thin cloud simulation method based on Perlin noise, domain warping, and atmospheric scattering model was proposed to construct a high-quality declouding dataset containing thin clouds of multiple sizes and shapes, which can improve the performance of DecloudNet on different kinds of thin clouds.

The remaining of this article is organized as follows. Section II describes the structure of DecloudNet, the details of cross-block consistency supervision, and the simulation methods for the four common thin clouds. The experimental settings and experimental results are given in Section III. The performance of DecloudNet and the existing state-of-the-art (SOTA) methods are compared on images from six different satellites. Section IV concludes this article.

II. Method

A. Cross-Patch Consistency Supervision

The existing dehazing methods have severe cross-patch color differences in the results when processing wide-swath images. This is due to the huge size of the wide-swath image, which needs to be cropped into patches before being fed into the network. The loss function of the existing method only focuses on the cloud removal effect of each patch fed into the network. No loss function is designed for the consistency of the cloud removal effect of each patch of the wide-swath image. To address this problem, we propose cross-patch consistency supervision. First, the cloud cover in the input image was made to vary by changing the image region input to the network. Second, the in-patch declouding loss (IPD Loss) was calculated between the declouding results and the ground-truth image, which was used to evaluate the thin cloud removal performance of the network. Third, the cross-patch consistency loss (CPC Loss) was computed between the same regions of the declouding results of different input images. Finally, the losses were summed up and the network parameters were updated.

The structure of cross-patch consistency supervision is shown in Fig. 2. For an image I input to DecloudNet, cut two small images I_1 and I_2 from it. Both I_1 and I_2 were fed into DecloudNet for processing to get two output images O_1 and O_2

$$O_1 = F_{\text{DecloudNet}}(I_1) \tag{1}$$

$$O_2 = F_{\text{DecloudNet}}(I_2). \tag{2}$$

Due to the change in the image area being fed into the network, the sizes and shapes of clouds in the image changed, resulting in a severe color difference between the two output images O_1 and O_2. As with most deep networks, the IPD Loss, which evaluated the declouding performance, was computed between O_1, O_2, and the ground truth GT_1 and GT_2

$$\text{Loss}_{I1} = F_{\text{IPDLoss}}(O_1, \text{GT}_1) \tag{3}$$

$$\text{Loss}_{I2} = F_{\text{IPDLoss}}(O_2, \text{GT}_2). \tag{4}$$

In addition, to resolve the severe color difference between the two output images, the CPC Loss function was designed. The size of the input image is set to 300×300 due to the limitation of GPU memory. S_1 and S_2 are the overlapping parts of O_1 and O_2, respectively. If the sizes of S_1 and S_2 are too large, there is too much overlap between O_1 and O_2. This leads to the inconsistency of the thin cloud removal of O_1 and O_2 not being significant enough. Thus, it is difficult to calculate the CPC Loss. If the sizes of S_1 and S_2 are too small, the overlap area of O_1 and O_2 is too small. The CPC Loss value calculated from the overlap area is too small. Thereby, it is difficult for the CPC Loss to play a supervision role for DecloudNet training. We varied the sizes of S_1 and S_2 in the range from 100×100 to 200×200 and train DecloudNet for several times. DecloudNet performs best when the sizes of S_1 and S_2 are 160×160 to 180×180. CPC Loss was computed between S_1 and S_2

$$\text{Loss}_C = F_{\text{CPCLoss}}(S_1, S_2). \tag{5}$$

Loss_{I1}, Loss_{I2} and Loss_C were added together as the total loss for network training

$$\text{Loss} = \lambda_1 \text{Loss}_{I1} + \lambda_2 \text{Loss}_{I2} + \lambda_3 \text{Loss}_C. \tag{6}$$

Loss_{I1}, Loss_{I2}, and Loss_C are all $L1$ Loss. The weights in (6) were chosen based on the values of the IPD Loss and the CPC Loss at the very beginning of DecloudNet training. In steps 0–2000 of training, the average IPD Loss was about 0.33 and the average CPC Loss was about 0.12. We made CPC Loss and IPD Loss equally important in DecloudNet training. Therefore, let the value of λ_3 be six times the value of λ_1 and λ_2. λ_1 and λ_2 took equal value. Change the values of λ_1, λ_2, and λ_3 for multiple training. DecloudNet performed best when the ratio of λ_3–λ_1 was between 4 and 6. For first training, the

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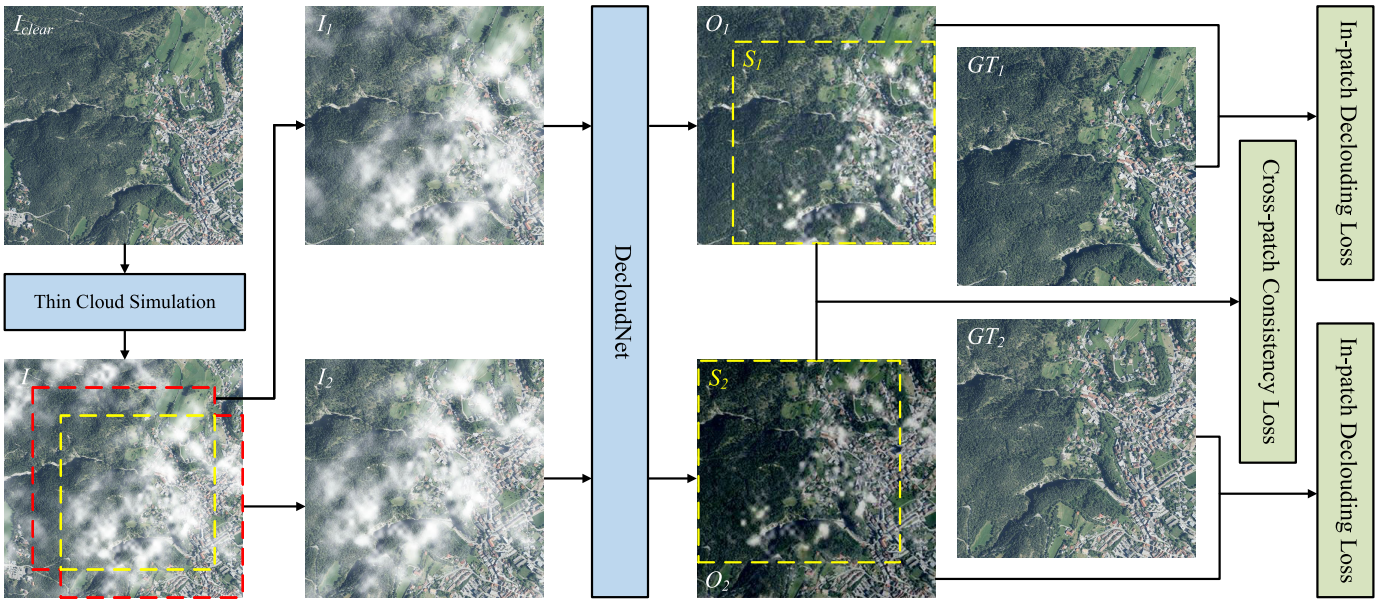


Fig. 2. Basic idea of cross-patch consistency supervision. By designing CPC Loss between partial overlapped patches, DecloudNet can effectively remove color difference between declouding results. The clear image is from the Inria Aerial Image Labeling dataset [38].

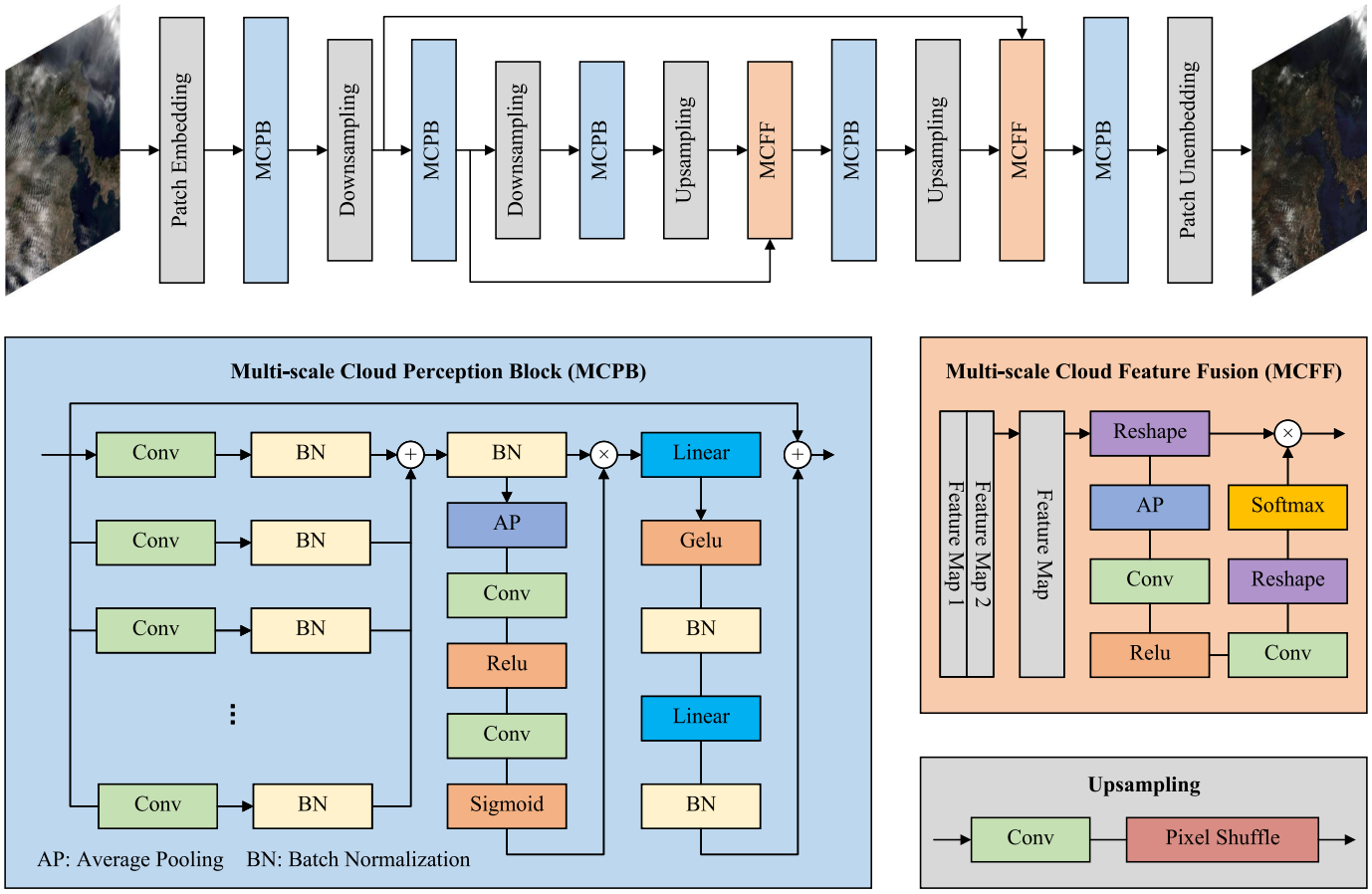


Fig. 3. DecloudNet was a five-stage U-Net composed of multiple MCPBs. The MCPB contained multiscaled convolutional kernels to perceive cloud features of different sizes and shapes. MCFF was used to fuse feature maps of different sizes.

initial values of λ_1, λ_2, and λ_3 can be made to be 0.8, 0.8, and 4.0. Minor adjustments can be made according to the decrease in IPD Loss and CPC Loss.

B. DecloudNet

To enhance the feature extraction capability of the network for clouds of multiple sizes and shapes in remote sensing

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TABLE I
CONVOLUTIONAL LAYER PARAMETERS FOR EACH MCPB

MCPB	Layers	Kernel Size
1	3	3, 3, 3
2	15	[5, 7, 3, 3, 3] × 3
3	63	[5, 7, 3, 3, 3, 3, 3] × 9
4	15	[5, 7, 3, 3, 3] × 3
5	3	3, 3, 3

images, we designed an MCPB consisting of convolutional layers of multiple kernel sizes and a squeeze-and-excitation net (SENet) [39] structure. Each MCPB module consisted of multiple parallel convolution-batch normalization, SENet, fully connected layers, and global residual connections. The structure of DecloudNet is shown in Fig. 3.

DecloudNet was a five-phase U-Net structured network containing five MCPBs. We used a small number of convolutional layers with large kernel sizes to rapidly increase the network receptive field in the leaning MCPB. The third MCPB contained a large number of small-sized convolutional kernels and a variety of different-sized convolutional kernels. A large number of small-size convolutional kernels were used to deepen the network and improve the feature dimension, and multiple different-sized convolutional kernels were used to perceive clouds of different sizes and shapes. The convolution kernels of the last two MCPBs were mirror-symmetric to the first two. This part of the network structure can be formulated as

$$I_1 = \mathrm{Add}\left(F^n_{\mathrm{BN}}\left(F^n_{\mathrm{conv}}(I)\right)\right) \tag{7}$$

where F^n_{conv} was the nth convolutional layer, and F^n_{BN} was the nth batch normalization. Add meant add by element. The number and size of convolutional layers in each MCPB are shown below. As seen in Table I, the first and fifth MCPBs have the same convolutional layers, as do the second and fourth MCPBs. In the second, third, and fourth MCPBs, there are three types of convolutional layers of size 3 × 3, 5 × 5, and 7 × 7. There are three groups of convolutional layers in the second and fourth MCPBs. Each group consists of one convolutional layer of size 5 × 5, one convolutional layer of size 7 × 7, and three convolutional layers of size 3 × 3. There are nine groups of convolutional layers in the third MCPB. Each group includes one convolutional layer of size 5 × 5, one convolutional layer of size 7 × 7, and five convolutional layers of size 3 × 3.

Clouds in multispectral images had different pixel values in different channels, so an SENet was introduced to give the network the ability to learn adaptive weights for different channels. The SE module modeled the relationship between channels by introducing a squeeze operation and an excitation operation. In the squeeze phase, it compressed the output feature map of the convolutional layer into a feature vector through a global average pooling operation. Then, in the excitation phase, it learned to generate a weight vector of the channels using a fully connected layer and a nonlinear activation function. This weight vector was applied to each channel on the original feature map to weight the features of different channels. In this way, the SE module was able to adaptively learn the importance of each channel and adjust the channel contributions in the feature map weighted according to the needs of the task. This attention mechanism helped the network to better focus on the important feature channels, thus improving the model performance.

In SE module, the feature map was first fed into the BN layer and the average pooling layer. In SE module, the feature map was first fed into the BN layer and the average pooling layer, changing its length and width to 1. Subsequently, the feature dimensions of the vectors were adjusted using two convolutional layers. This process can be formulated as

$$I_s = \mathrm{Sigmoid}(\mathrm{Conv}(\mathrm{Relu}(\mathrm{Conv}(\mathrm{AP}(\mathrm{BN}(I_1)))))) \tag{8}$$

where I_s was a vector of shape $B \times C \times 1 \times 1$. Multiply I_s and I_1 by channel to get the output I_2 of the SE module

$$I_2 = I_s \times I_1. \tag{9}$$

Subsequently, the output I_M of the MCPB module was obtained after processing of two fully connected layers, Gelu activation function, and two BN layers

$$I_M = \mathrm{BN}(\mathrm{Linear}(\mathrm{BN}(\mathrm{Gelu}(\mathrm{Linear}(I_2))))). \tag{10}$$

In addition to MCPB, DecloudNet included a multiscale cloud feature fusion (MCFF) modified from selective kernel (SK) Fusion [40], up- and downsampling. MCFF was used to fuse feature maps of different stages of the network which have different scales. The two feature maps being fed into MCFF were first connected by channel

$$I_C = \mathrm{Concat}(I_{F1}, I_{F2}). \tag{11}$$

Subsequently, I_C was reshaped to $B \times 2 \times C \times H \times W$ and was processed through average pooling, convolutional layers, and Relu activation functions, after which it was again reshaped to $B \times 2 \times C \times 1 \times 1$

$$I_{R1} = \mathrm{Reshape}(I_C) \tag{12}$$

$$I_{R2} = \mathrm{Reshape}(\mathrm{Conv}(\mathrm{Relu}(\mathrm{Conv}(\mathrm{AP}(I_R))))) \tag{13}$$

I_{R1} and I_{R2} were multiplied by channel, and the output I_F of MCFF was obtained after compressing the second dimension

$$I_F = \mathrm{Reshape}(I_{R1} \times I_{R2}). \tag{14}$$

When the remote sensing image containing clouds was fed into DecloudNet, it was processed by patch embedding and the first MCPB module

$$I_{M1} = F_{\mathrm{MCPB1}}(F_{\mathrm{PE}}(I)). \tag{15}$$

After that, I_{M1} was downsampled and an MCPB processed, repeated twice

$$I_{M2} = F_{\mathrm{MCPB2}}(\mathrm{Downsampling}(I_{M1})) \tag{16}$$

$$I_{M3} = F_{\mathrm{MCPB3}}(\mathrm{Downsampling}(I_{M2})). \tag{17}$$

Subsequently, the feature maps were sequentially upsampled, MCFF and MCPB processed, and repeated twice

$$I_{M4} = F_{\mathrm{MCFF1}}(\mathrm{Upsampling}(I_{M3}), I_{M2}) \tag{18}$$

$$I_{M5} = F_{\mathrm{MCFF2}}(\mathrm{Upsampling}(I_{M4}), I_{M1}). \tag{19}$$

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Finally, the output I_{decloud} was obtained after patch unembedding module

$$I_{\text{decloud}} = F_{\text{PU}}(I_{M5}). \tag{20}$$

C. Thin Cloud Simulation Method

High-quality thin cloud removal samples are the basis for training thin cloud removal models. A pair of ideal thin cloud removal samples are cloud-free and cloudy images of the same region at the same time. However, due to the influence of light, weather, and other factors, it is difficult to obtain ideal thin cloud removal samples in practical research. Most existing datasets use algorithms to generate simulated thin cloud removal samples. Most of the existing dehazing datasets contain only images that are covered by simulated fog. Models trained on these datasets have had difficulty in handling thin clouds with rich and variable sizes and shapes. In addition, most of these images are taken from a horizontal view, which are significantly different from satellite images taken from a top view.

To solve these problems, a thin cloud simulation method based on Perlin Noise, Domain Warping and Atmospheric Scattering Model is proposed in this article. To better generate simulated thin clouds, this article first classifies common thin clouds in remote sensing images. From a meteorological point of view, clouds can be categorized into three major cloud families and ten cloud genera. Among them, thin clouds mainly contain the following 11 cloud categories: Cumulus humilis (Cu hum) and Cumulus fractus (Cu fra) in the genus Cumulus (Cu), Stratus nebulosus (St neb) in the genus Stratus (St), thin Altostratus (As) and Altocumulus floccus (Ac flo) in Altocumulus (Ac), Cirrostratus fibratus (Cs fib) and Cirrostratus nebulosus (Cs neb) in Cirrostratus (Cs), and Cirrocumulus castellanus (Cc cas), Cirrocumulus floccus (Cc flo), and Cirrocumulus stratiformis (Cc str) in Cirrocumulus (Cc) and Cirrus (Ci).

These thin clouds are classified into several categories from a meteorological point of view. However, they can be combined into four categories based on how they look from a satellite view. First, Cu hum, Cu fra, Ac flo, and Cc flo can be categorized as popcorn-like thin cloud. Second, Cc cas and Cc str can be categorized as wave-like thin cloud. Third, Cs fib and Ci can be categorized as cotton-like thin cloud. Finally, St neb, As, and Cs neb can be categorized as fog-like thin cloud. The classification results are shown in Table II. The four kinds of thin clouds are shown in Fig. 4.

These four kinds of thin clouds have their own characteristics in terms of size and shape. Therefore, thin clouds were simulated using three methods in this article. For popcorn-like thin clouds, a Perlin-noise-based method modified from [41] was used for simulation. The interpolation method for Perlin noise is cubic spline interpolation. Noise images at different scales were generated by multiple interpolation. These noise images were summed to obtain the simulated thin cloud

$$I = \sum_{i=1}^{n} iCi + \left(1 - \sum_{i=1}^{n} iCi\right) I_{\text{clear}} \tag{21}$$

TABLE II

ELEVEN KINDS OF THIN CLOUDS HAVE BEEN CATEGORIZED INTO FOUR KINDS BASED ON HOW THEY LOOK FROM A SATELLITE VIEW

Classification from a remote sensing perspective	Classification from a meteorological perspective
Popcorn-like thin cloud	Cu hum, Cu fra, Ac flo, Cc flo
Wave-like thin cloud	Cc cas, Cc str
Cotton-like thin cloud	Cs fib, Ci
Fog-like thin cloud	St neb, As, Cs neb

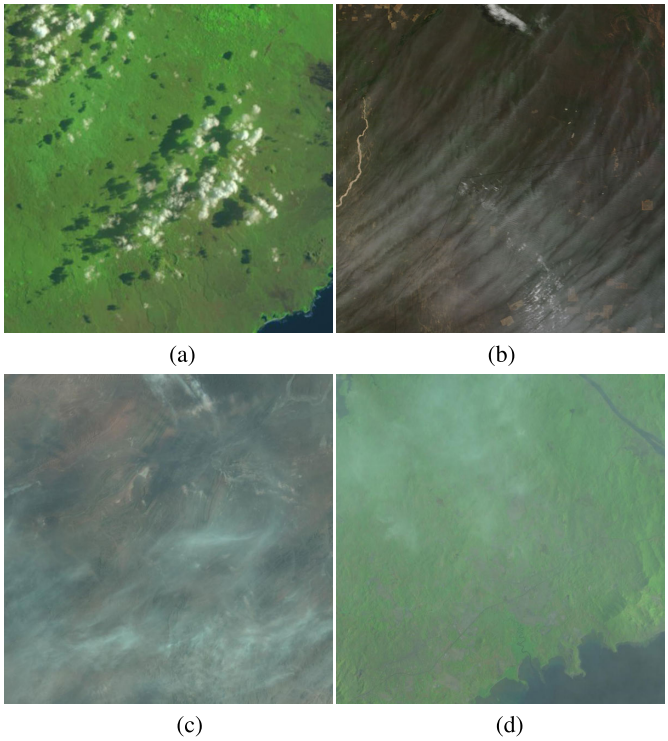


Fig. 4. Typical images of four kinds of thin cloud in remote sensing images. (a) Popcorn-like thin cloud. (b) Wave-like thin cloud. (c) Cotton-like thin cloud. (d) Fog-like thin cloud.

where C_i is the noise image at different scales obtained by multiple interpolation, and I_{clear} is the cloud-free image.

For wave-like thin cloud and cotton-like thin cloud, domain warping [42] based on fractal Brownian motion (fBM) was used to simulate curved cloud. Let p be a pixel position in the picture space domain. Warping means distorting the position p before generating noise on the picture. Position p was distorted by fBM twice first, and then the noise was generated using fBM. This process can be formulated as follows:

$$I_n = f_{\text{fBM}}(p + f_{\text{fBM}}(p + f_{\text{fBM}}(p))) \tag{22}$$

where I_n is the generated noise images. Simulated thin cloud was obtained after color and brightness adjustments to noise image. The shape of the generated simulated thin cloud is determined by the distorted position p. Domain warping uses a rotation matrix to distort the position. The shape of the generated simulated thin clouds can be controlled by changing the elements of the matrix. The rotation matrix used in

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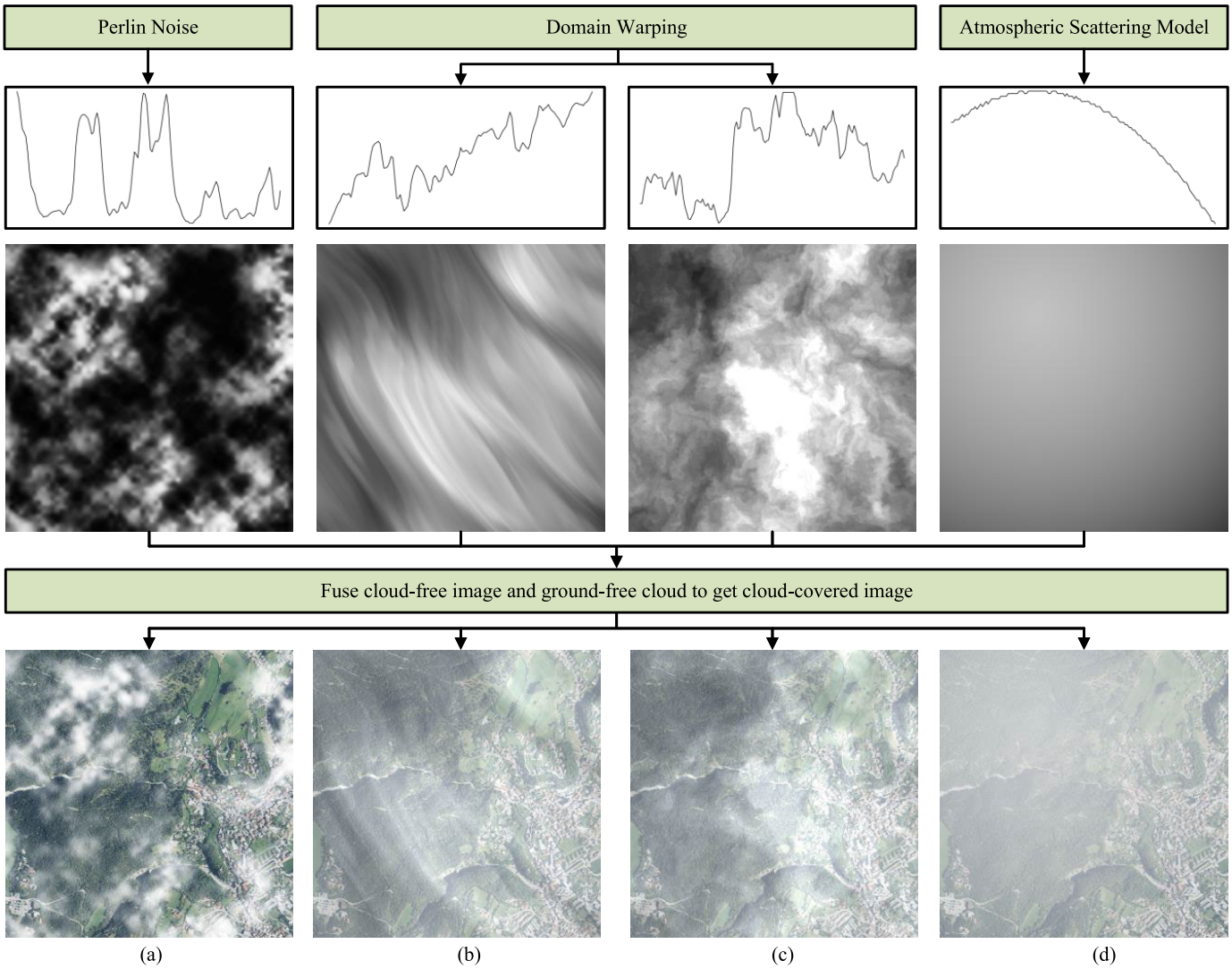


Fig. 5. Thin cloud removal sample generation method based on Perlin noise, domain warping, and atmospheric scattering model. Perlin noise was used to generate popcorn-like thin cloud. Wave-like thin cloud and cotton-like thin cloud were generated using domain warping based on fBM. Fog-like thin cloud were generated based on the atmospheric scattering model. (a) Popcorn-like thin cloud. (b) Wave-like thin cloud. (c) Cotton-like thin cloud. (d) Fog-like thin cloud.

generating wave-like thin clouds is as follows:

$$\mathbf{R} = \begin{pmatrix} \cos(1) & \sin(1) \\ 0 & 1 \end{pmatrix}. \tag{23}$$

The rotation matrix used in generating cotton-like thin clouds is as follows:

$$\mathbf{R} = \begin{pmatrix} 1 & \sin(1) \\ -\sin(1) & 1 \end{pmatrix}. \tag{24}$$

For fog-like thin cloud, the simulation was performed using atmospheric scattering model [43]. The atmospheric scattering model can be formulated as follows:

$$I_f(x) = J(x)t(x) + A_\infty(1 - t(x)) \tag{25}$$

$$t(x) = e^{-\beta d(x)} \tag{26}$$

where $I_f(x)$ is the foggy image, $J(x)$ is the reflected light from the object, $t(x)$ is the atmospheric transmissivity, A_∞ is the airlight at infinity, β is the scattering coefficient, and $d(x)$ is the depth of field. Let β be 0.04, A_∞ be 0.9, and $d(x)$ be adjusted according to the image size to get the simulated fog-like thin cloud. The simulation results of these four kinds of thin clouds are shown in Fig. 5.

III. Experiments

A. Dataset and Settings

To address the problem that the existing remote sensing image cloud removal dataset contains insufficient variety of thin clouds, we produced a new Remote Sensing Declouding (RSD) dataset. The RSD dataset contains images from six different satellites: Quick Bird (QB), GaoFen-2 (GF-2), World View-2 (WV-2), GaoFen-1 Wide Field of View (GF-1 WFV), GaoFen-1 (GF-1), and Earth-Observing-1 (EO-1). The high spatial resolution images from the first three satellites were used to enable network learning to recover details of ground objects covered by thin clouds. The latter three satellites have a wide imaging range and were used to enable network learning to maintain cross-patch consistency when processing wide-swath images. The RSD dataset contains a variety of images

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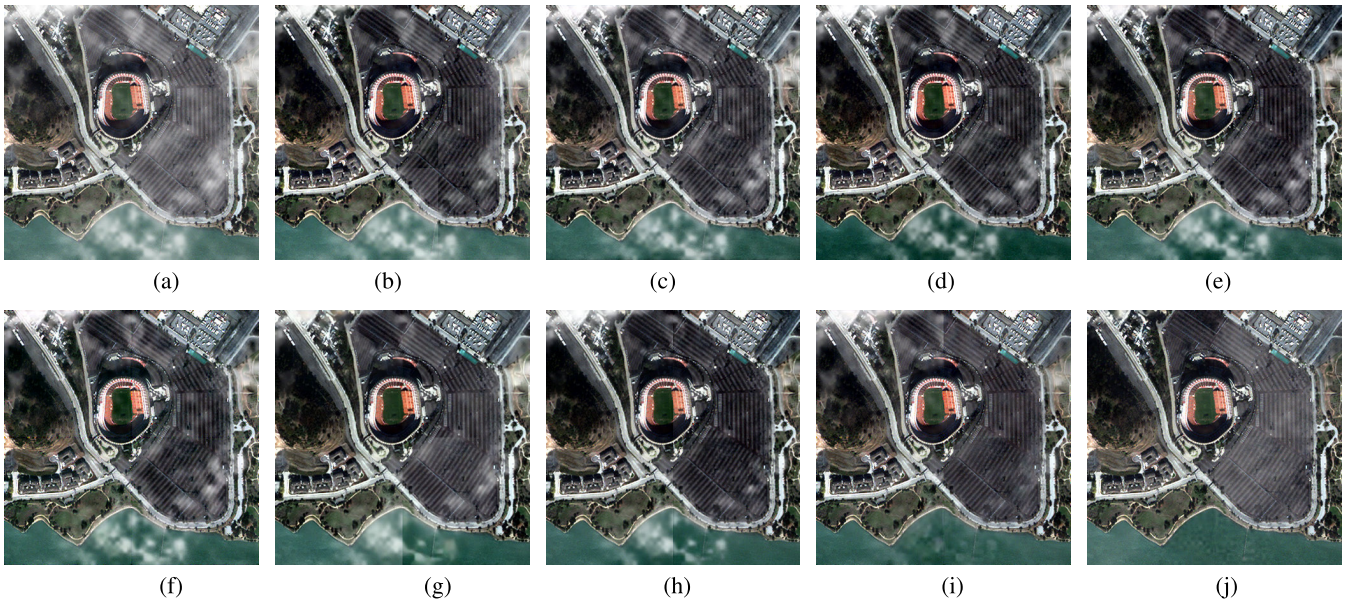


Fig. 6. Comparison of different declouding methods on the simulated popcorn-like thin cloud. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

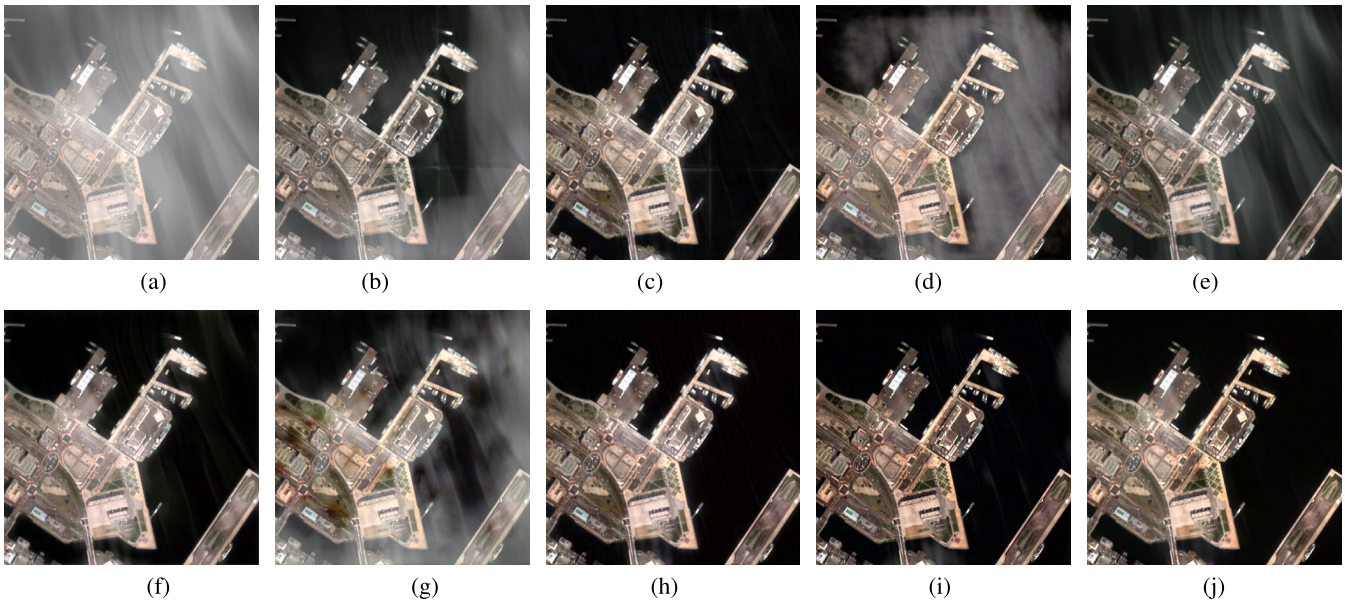


Fig. 7. Comparison of different declouding methods on the simulated wave-like thin cloud. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

of real thin cloud cover scenes and simulated images of four common types of thin clouds. This facilitates the network to fully learn the features of various thin clouds and improve the declouding performance.

The RSD dataset consists of a training set and a test set. The training set contains 5000 pairs of cloudy and cloud-free images. Among them, 2700 pairs of images are from QB, GF-2, and WV-2 satellites. By the thin cloud simulation method proposed in this article, four common types of thin clouds were added to these images: popcorn-like, cotton-like, wave-like, and fog-like thin clouds. The remaining 2600 pairs of images are from the RICE [35] and RSHaze [40] datasets. These images contain multiple scenes of real cloud cover. All the images in the training set were cropped to 400 $\times$ 400. The test set contains 430 test images. Of these, 340 images are from the QB, GF-2, and WV-2 satellites. These images contain four common simulated thin clouds. They were used to test the network's ability to remove local thin clouds and to reconstruct details of ground objects. The remaining 90 images are from the GF-1 WFV, GF-1, and EO-1 satellites. The images range in size from 1000 $\times$ 1000 to 10 000 $\times$ 10 000 and contain real clouds of various sizes and shapes. They were used to test the network's ability to remove clouds from wide-swath images and maintain cross-patch consistency. Both our method and the compared methods were trained and tested on an Ubuntu 20.04 server with an RTX-4090 GPU. The server has a cuda version of 12.0 and a PyTorch version of 1.13.1.

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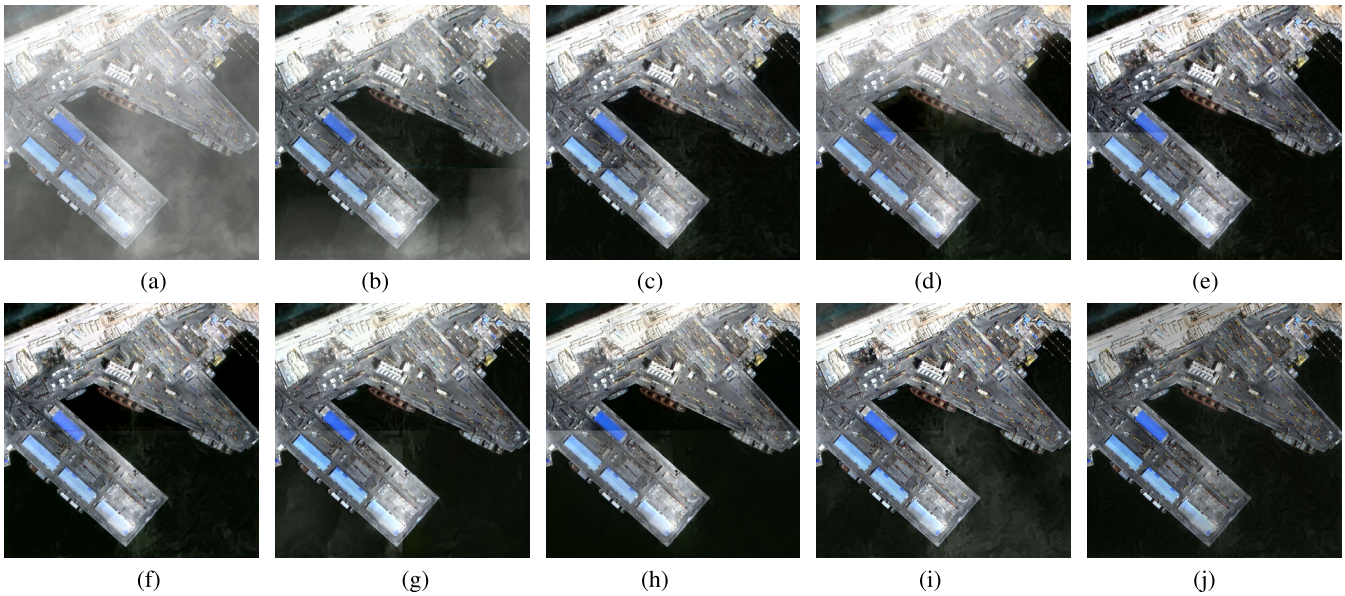


Fig. 8. Comparison of different declouding methods on the simulated cotton-like thin clouds image. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

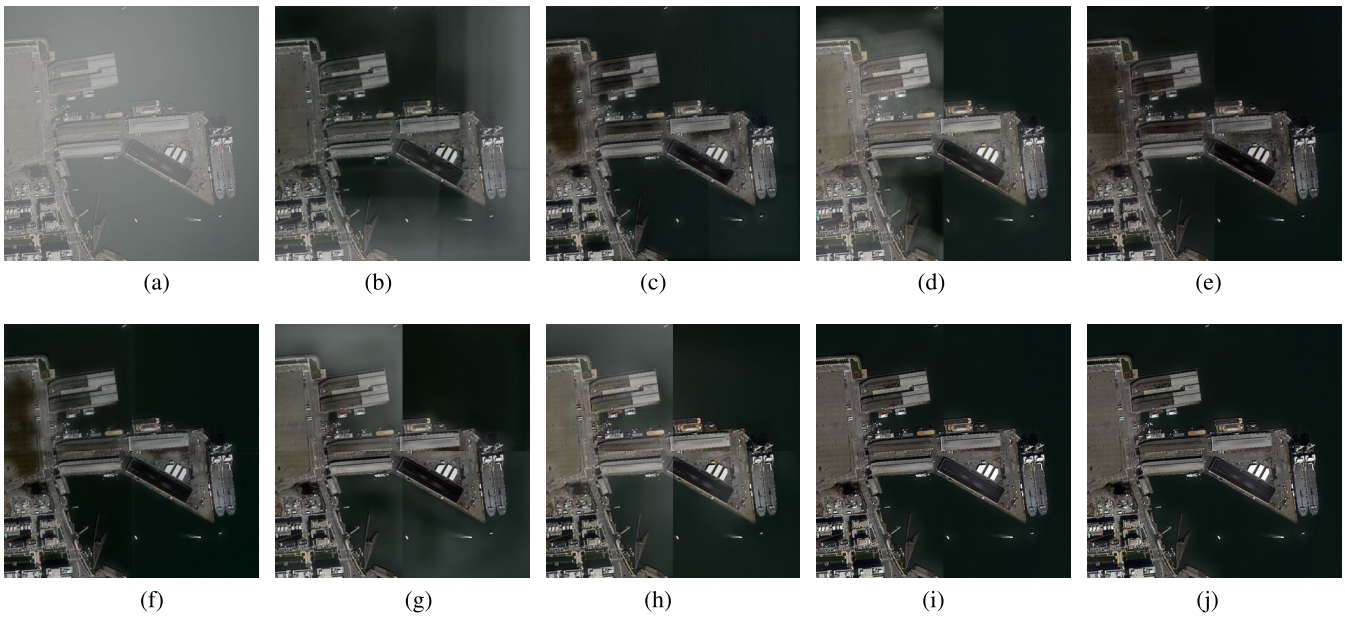


Fig. 9. Comparison of different declouding methods on the simulated fog-like thin cloud. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

B. Comparison With State-of-the-Art Methods

We compared our method with eight SOTA methods: Trinity dehazing network (TrinityNet) [33], first-coarse-then-fine two-stage dehazing neural network (FCTFNet) [34], FFANet [24], gating U-Net (gUNet) [44], curricular contrastive regularization for physics-aware single image dehazing network (C2PNet) [45], detail-enhanced attention dehazing network (DEANet) [46], Mixdehazenet [26], and vision transformers for single image dehazing (Dehazeformer) [40]. Among them, TrinityNet and FCTFNet were designed for remote sensing images, and the other methods were designed for natural images. On one hand, we tested the localized declouding performance of the network on high spatial resolution images from the QB, GF-2, and WV-2 satellites. These images contain four types of simulated thin clouds. On the other hand, we tested the global declouding performance of the network on wide-swath images from the GF-1 WFV, GF-1, and EO-1 satellites. These images contain a variety of real clouds.

Declouding results of DecloudNet and comparison methods for the simulated clouds are shown in Figs. 6–9. In terms of the type of thin clouds, cotton and foggy thin clouds are less difficult to process. More than half of the methods can remove them well. Popcorn-like and wave-like thin clouds are more difficult to process, especially popcorn-like thin clouds, only Dehazeformer and DecloudNet proposed in this

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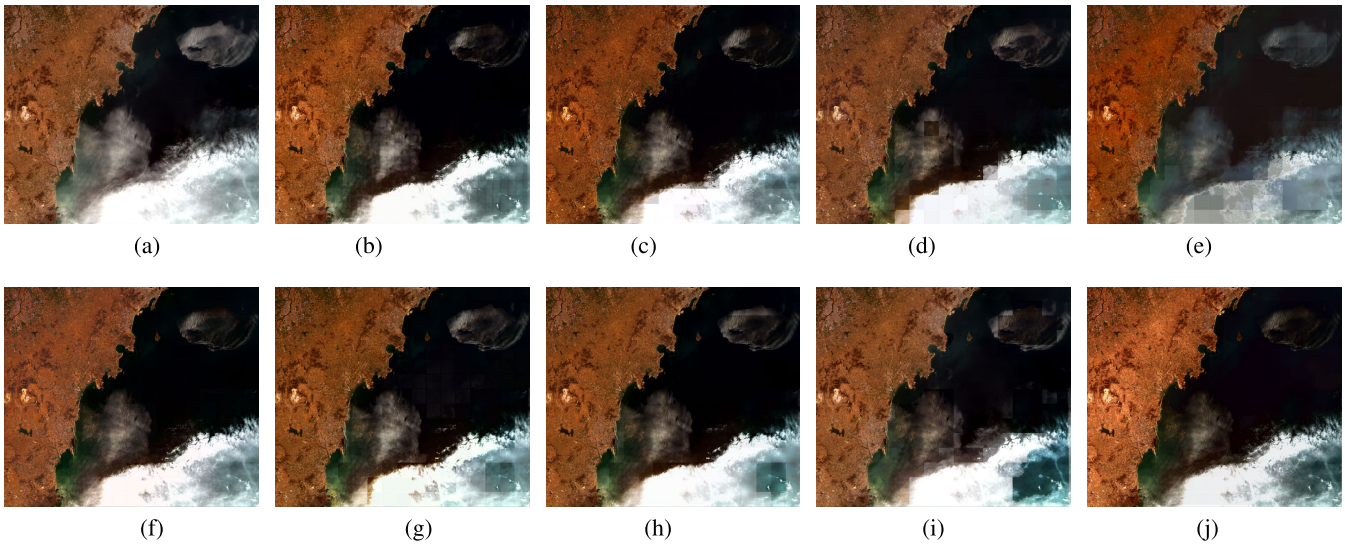


Fig. 10. Comparison of different declouding methods on the GF-1-WFV image with real wave-like, cotton-like, and fog-like thin clouds. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

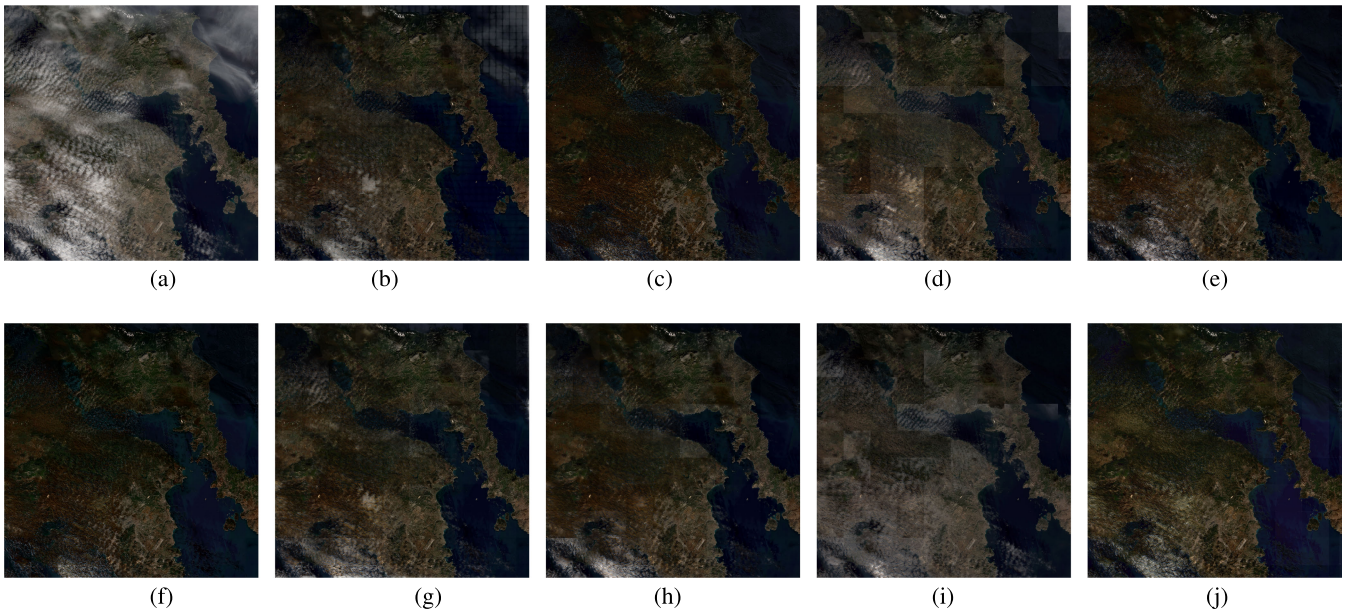


Fig. 11. Comparison of different declouding methods on the GF-1 image with real popcorn-like, wave-like, and cotton-like thin clouds. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

article can remove them from the images. This is because popcorn-like thin clouds are cumulus clouds, which have complex and variable shapes and are thicker than other thin clouds. For different declouding methods, most of the methods can remove only one or two types of thin clouds, and a few methods such as C2PNet and Dehazeformer can remove three types of thin clouds. The DecloudNet proposed in this article can effectively remove all four types of thin clouds. In addition, despite the small size of the test images, there were cross-patch color difference in the declouding results for each compared method. The performance of each declouding method was evaluated by peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) metrics as shown in Table III. DecloudNet outperformed the compared methods in all but two figures.

Declouding results of DecloudNet and comparison methods for real clouds are shown in Figs. 10–12. These images are wide-swath images and are huge in size. As can be seen from Figs. 10–12, the cross-patch color difference in the results of the compared method increased significantly as the image size increases. In this article, DecloudNet was trained under cross-patch consistency supervision. DecloudNet effectively eliminates cross-patch color difference while maintaining good thin cloud removal. This allows most GPUs to handle wide-swath images of huge dimensions without being limited by the size of the graphics memory. The performance

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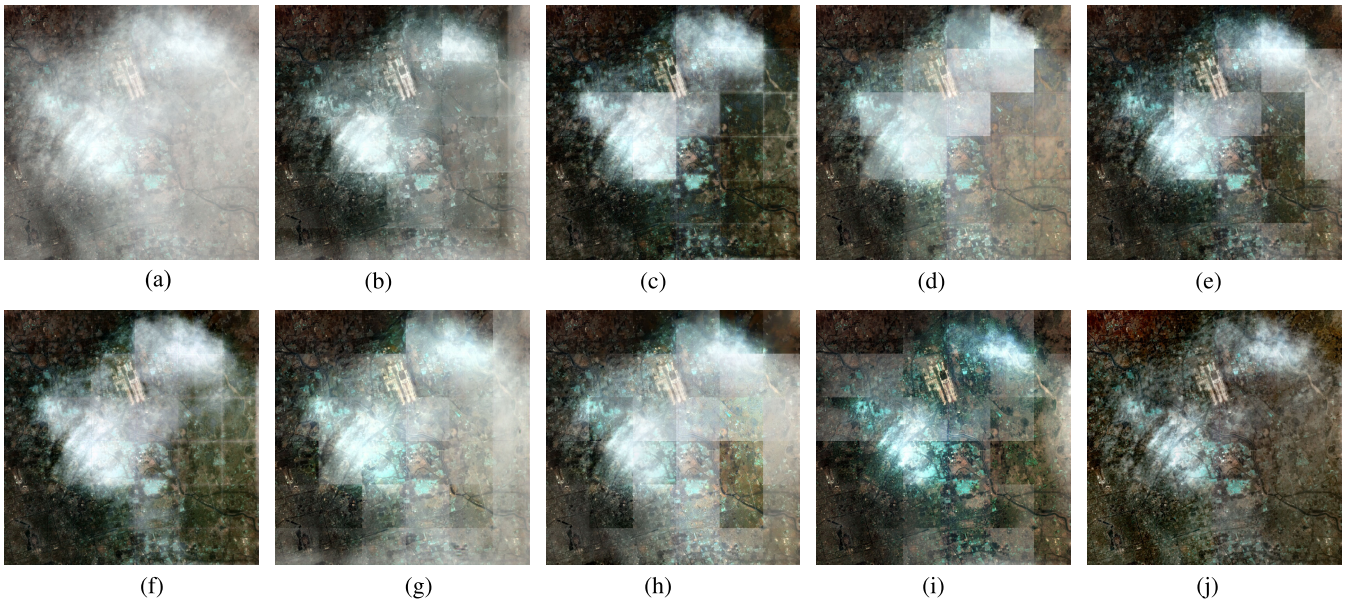


Fig. 12. Comparison of different declouding methods on the EO-1 image with real wave-like, cotton-like, and fog-like thin clouds. (a) Input. (b) TrinityNet [33]. (c) FCTFNet [34]. (d) FFANet [24]. (e) gUNet [44]. (f) C2PNet [45]. (g) DEANet [46]. (h) MixDehazeNet [26]. (i) Dehazeformer [40]. (j) DecloudNet (Ours).

TABLE III
COMPARISON OF DIFFERENT DECLOUDING METHODS ON FOUR TYPES OF SIMULATED THIN CLOUD-COVERED IMAGES FROM QB, GF-2, AND WV-2 SATELLITES

Satellite	Method	Parameters	Popcorn-like Cloud		Cotton-like Cloud		Wave-like Cloud		Fog-like Cloud	
			PSNR(dB)↑	SSIM↑	PSNR(dB)↑	SSIM↑	PSNR(dB)↑	SSIM↑	PSNR(dB)↑	SSIM↑
QB	TrinityNet [33]	20.24M	16.23	0.7417	17.12	0.8077	16.42	0.7782	18.11	0.8121
	FCTFNet [34]	0.164M	14.93	0.7185	19.28	0.8154	16.92	0.8073	19.56	0.8158
	FFANet [24]	2.192M	16.30	0.7224	19.35	0.7889	17.02	0.7617	21.77	0.8300
	gUNet [44]	1.477M	15.41	0.7168	20.22	0.8418	17.79	0.8414	19.50	0.8320
	C2PNet [45]	3.135M	15.19	0.6835	20.79	0.7973	18.87	0.8121	19.03	0.8134
	DEANet [46]	7.789M	18.73	0.7984	20.94	0.8248	19.02	0.8378	22.16	0.8596
	MixDehazeNet [26]	3.165M	17.42	0.7407	22.18	0.8501	20.50	0.8492	22.07	0.8295
	Dehazeformer [40]	2.517M	20.82	0.8334	22.48	0.8315	20.94	0.8843	21.17	0.8566
	DecloudNet(Ours)	6.599M	**21.24**	**0.8485**	**22.50**	**0.8634**	**21.15**	**0.9150**	**22.20**	**0.8776**
GF-2	TrinityNet [33]	20.24M	15.27	0.7928	17.14	0.7839	15.72	0.7173	19.11	0.8563
	FCTFNet [34]	0.164M	14.72	0.7677	20.87	0.8409	16.19	0.8089	20.24	0.8389
	FFANet [24]	2.192M	16.29	0.7678	17.88	0.8412	15.77	0.8082	21.56	0.8534
	gUNet [44]	1.477M	14.40	0.7644	18.50	0.8626	16.91	0.8457	20.58	0.8557
	C2PNet [45]	3.135M	14.81	0.7574	19.15	0.8628	17.41	0.8476	20.41	0.8488
	DEANet [46]	7.789M	17.40	0.8225	19.68	0.8733	17.89	0.8557	21.99	0.9065
	MixDehazeNet [26]	3.165M	16.89	0.8114	20.70	0.8804	19.31	0.8708	21.75	0.8848
	Dehazeformer [40]	2.517M	20.75	0.8217	21.64	**0.9060**	19.80	0.8984	20.61	0.8964
	DecloudNet(Ours)	6.599M	**21.06**	**0.8241**	**22.49**	0.8974	**20.36**	**0.9096**	**22.64**	**0.9076**
WV-2	TrinityNet [33]	20.24M	18.61	0.8306	16.42	0.7623	17.27	0.7539	19.43	0.7377
	FCTFNet [34]	0.164M	17.50	0.7961	17.46	0.7922	19.35	0.8156	21.29	0.8733
	FFANet [24]	2.192M	15.11	0.7047	16.56	0.7884	15.37	0.7945	20.20	0.7767
	gUNet [44]	1.477M	14.96	0.7028	20.22	0.8529	19.48	0.8876	20.61	0.7860
	C2PNet [45]	3.135M	14.82	0.6711	19.74	0.8172	19.18	0.8536	20.58	0.7714
	DEANet [46]	7.789M	18.19	0.7669	19.67	0.8281	18.24	0.8355	21.54	0.8078
	MixDehazeNet [26]	3.165M	16.62	0.7335	20.03	0.8377	19.27	0.8606	22.25	0.8042
	Dehazeformer [40]	2.517M	19.78	0.8119	21.73	0.8865	21.24	**0.9121**	24.29	0.8992
	DecloudNet(Ours)	6.599M	**20.51**	**0.8310**	**22.52**	**0.9064**	**22.11**	0.9055	**24.33**	**0.9004**

of DecloudNet and compared methods on real cloudy images was evaluated by the blind/referenceless image spatial quality evaluator (BRISQUE) [47], natural image quality evaluator (NIQE) [48], and perception-based image quality evaluator (PIQE) [49] metrics as shown in Table IV. DecloudNet outperformed the compared methods in all but one figure.

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TABLE IV
COMPARISON OF DIFFERENT DECLOUDING METHODS ON REAL CLOUD-COVERED WIDE-SWATH IMAGES FROM GF-1 WFV, GF-1, AND EO-1 SATELLITES

Method	Parameters	GF-1-WFV			GF-1			EO-1		
		BRISQUE↓	NIQE↓	PIQE↓	BRISQUE↓	NIQE↓	PIQE↓	BRISQUE↓	NIQE↓	PIQE↓
TrinityNet [33]	20.24M	78.36	10.93	10.35	30.87	8.33	13.41	37.28	14.60	14.40
FCTFNet [34]	0.1635M	74.65	11.34	9.12	31.24	8.46	12.17	37.56	15.86	13.48
FFANet [24]	2.192M	78.74	9.73	8.06	33.41	11.13	13.92	37.97	14.52	14.82
gUNet [44]	1.477M	92.58	10.45	7.32	29.11	8.14	11.33	39.62	13.74	9.21
C2PNet [45]	3.135M	85.80	9.68	8.46	32.72	10.45	11.89	39.71	16.29	8.52
DEANet [46]	7.789M	79.47	8.83	8.97	**27.65**	8.83	12.44	38.50	14.70	11.65
MixDehazeNet [26]	3.165M	79.49	8.56	9.07	32.03	7.60	11.24	39.63	12.62	13.11
Dehazeformer [40]	2.517M	73.12	10.25	8.49	32.44	9.37	14.24	41.20	19.02	12.18
DecloudNet(Ours)	6.599M	**70.83**	**8.08**	**7.10**	28.58	**7.02**	**11.12**	**35.44**	**12.12**	**8.23**

TABLE V
COMPARISON OF DIFFERENT DECLOUDNET ON SIMULATED THIN-CLOUD-COVERED IMAGES

Method	PSNR (dB)↑	SSIM↑
DecloudNet-Baseline	14.79	0.6249
DecloudNet-R	16.94	0.7021
DecloudNet-RM	19.63	0.8274
DecloudNet-RMC	**21.93**	**0.8822**

TABLE VI
COMPARISON OF DIFFERENT DECLOUDNET ON REAL-CLOUD-COVERED IMAGES

Method	BRISQUE↓	NIQE↓	PIQE↓
DecloudNet-Baseline	74.75	14.33	13.62
DecloudNet-R	68.21	13.36	11.84
DecloudNet-RM	58.76	10.92	10.37
DecloudNet-RMC	**44.95**	**9.07**	**8.82**

C. Ablation Study

To verify the effectiveness of the MCPB, cross-patch consistency supervision, and RSD dataset proposed in this article for improving the performance of declouding, we conducted a series of ablation experiments. By replacing MCPB with convolutional kernels of 3×3 size, not applying CPC Loss and training DecloudNet only on the RICE and RSHaze datasets, we obtain DecloudNet-Baseline. First, DecloudNet-R was obtained by training DecloudNet-Baseline on the RSD dataset. Second, MCPB was added to DecloudNet-R to obtain DecloudNet-RM. Finally, the CPC Loss was applied to DecloudNet-RM to obtain DecloudNet-RMC.

The average PSNR and SSIM of the results of the four DecloudNets on the simulated cloud images are shown in Table V, and the BRISQUE, NIQE, and PIQE of the results on real cloud images are shown in Table VI. As seen in Table VI, MCPB, CPC Loss, and RSD dataset are all effective in improving DecloudNet's thin cloud removal performance.

IV. CONCLUSION

To address the shortcomings of existing methods and datasets for the task of thin cloud removal from wide-swath image, this article proposed DecloudNet composed of MCPBs, cross-patch consistency supervision, and RSD dataset. First, the MCPB containing multiple size convolutional kernels enhanced DecloudNet's ability to extract the features of thin clouds with different sizes and shapes. Second, IPD Loss and CPC Loss were used to supervise DecloudNet's training. The cross-patch color difference in the thin cloud removal results was reduced while maintaining good declouding performance. Finally, 11 kinds of thin clouds in meteorology were categorized into four kinds from the remote sensing perspective. The four kinds of thin clouds were simulated using methods based on Perlin noise, domain warping, and atmospheric scattering model to provide a high quality dataset for DecloudNet training. DecloudNet and compared methods were tested on thin cloud covered images from six different satellites: QB, GF-2, WV-2, GF-1 WFV, GF-1, and EO-1. Exhaustive experiments and ablations illustrate that DecloudNet removes both the simulated and real thin clouds with better performance than the existing SOTA methods. Moreover, DecloudNet allows most GPUs to handle wide-swath images of huge size without worrying about graphics memory limitation. DecloudNet has also been experimented with for processing wide-swath images with size as large as $100\,000 \times 30\,000$. There is still no cross-patch color difference in the results, but the thin cloud removal performance needs to be improved. In the future, research will be continued for wide-swath image thin cloud removal.

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